(c) Measures of dispersion		
2c.01	Calculate different measures of spread: range, quartiles, interquartile range (IQR), percentiles, **interpercentile range, interdecile range and standard deviation**.	**For example, 10th to 90th interpercentile range.** Any value of n may be expected, so that required bounds (e.g. quartiles) may or may not be values in the data set. Alternative methods will be acceptable provided that the method used is clear from the working. (e.g. if the median lies between two data values the arithmetic mean of these two values may be used.) **For standard deviation only the formulae for a set of values are given. Students will need to know how to apply these to grouped data, i.e.** **Standard deviation =** $\sqrt{\frac{\sum f(x-\bar{x})^2}{\sum f}}$ **or** $\sqrt{\frac{\sum fx^2}{\sum f}-\left(\frac{\sum fx}{\sum f}\right)^2}$
2c.02	Identify outliers by inspection **and using appropriate calculations.**	**Calculations are expected to be known:** **Small outlier is** **< LQ – 1.5 × IQR** **Large outlier is** **> UQ + 1.5 × IQR** **Or outlier is outside** $\mu \pm 3\sigma$

Higher tier content

What students need to learn:		Guidance
(c) Measures of dispersion *continued*		
2c.03	Comment on outliers with reference to the original data.	Know that outliers may be genuine unusual values or may be the result of errors in recording data. **Outlier boundaries may need to be calculated.**
2c.04	Compare different data sets using appropriate calculated or given measure of spread: range, interquartile range (IQR), percentiles **and standard deviation**.	An awareness of which measure is more appropriate to use is expected, e.g. selecting the appropriate values from those produced by statistical software.
2c.05	Use calculated or given median and interquartile range (IQR) **or interpercentile range or interdecile range or mean and standard deviation** to compare data samples and to compare sample data with population data.	The appropriate pairing of a measure of central tendency and a measure of dispersion is expected. (e.g. use of mean with IQR is not appropriate.)
2c.06	**Use calculated or given means and standard deviation to standardise and interpret data collected in two comparable samples.** **Formulae for standard deviation will be given in the formulae sheet.**	**Know how to standardise using these values:** **standardised score =** $\frac{x-\mu}{\sigma}$ **(Formula will not be given.)**